Worksheet 3D: Graphing Quadratics

About This Worksheet

This worksheet connects the algebra you've learned to visual understanding. You'll see how changing numbers in an equation changes the shape and position of a parabola.

GeoGebra Links: Throughout this worksheet, you'll find links to interactive GeoGebra applets. Open these in a browser to explore the concepts hands-on.

Part A: The Basic Parabola (8 problems)

The simplest quadratic function is:

$$f(x) = x^2$$

** Explore:** Basic Parabola *Type* `y = x^2` *into GeoGebra to see it.*

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1. Complete this table for $f(x) = x^2$:

x	-3	-2	-1	0	1	2	3
$f(x)$							

2. Plot these points on graph paper and connect them smoothly.

3. Answer these questions about $f(x) = x^2$: - (a) What is the lowest point on the graph? (This is called the **vertex**) - (b) Is the parabola symmetric? Around what line? - (c) Does the parabola open upward or downward? - (d) What is $f(0)$? This is the **y-intercept**.

4. The graph of $f(x) = x^2$ never goes below the x-axis. What does this tell you about: - (a) The solutions to $x^2 = 0$? - (b) The solutions to $x^2 = -1$?

5. Now consider $g(x) = -x^2$. Complete the table:

x	-3	-2	-1	0	1	2	3
$g(x)$							

6. How does $g(x) = -x^2$ compare to $f(x) = x^2$?

7. For $g(x) = -x^2$: - (a) What is the vertex? - (b) Does it open upward or downward? - (c) What is the maximum or minimum value?

8. Rule: If the coefficient of x^2 is positive, the parabola opens _____. If negative, it opens _____.

Part B: Vertical Stretch and Compression (8 problems)

** Explore:** Open GeoGebra and graph these together: - $y = x^2$ (red) - $y = 2x^2$ (blue) - $y = 0.5x^2$ (green) - $y = 3x^2$ (orange)

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9. Complete the table comparing $f(x) = x^2$ and $g(x) = 2x^2$:

x	-2	-1	0	1	2
$f(x) = x^2$					
$g(x) = 2x^2$					

10. How does multiplying by 2 affect the y-values?

11. Compare $f(x) = x^2$ and $h(x) = 0.5x^2$:

x	-2	-1	0	1	2
$f(x) = x^2$					
$h(x) = 0.5x^2$					

12. How does multiplying by 0.5 affect the y-values?

13. Which parabola is "narrower" (steeper): $y = x^2$ or $y = 3x^2$?

14. Which parabola is "wider" (flatter): $y = x^2$ or $y = 0.25x^2$?

15. Match each equation to its description:

Equation	Description
$y = 5x^2$	A. Same as $y = x^2$ but flipped
$y = 0.1x^2$	B. Very narrow, opens up
$y = -x^2$	C. Very wide, opens up
$y = -3x^2$	D. Narrow, opens down

16. Rule: The coefficient a in $y = ax^2$ affects: - **Sign of a:** Determines if parabola opens _____ or _____ - **Size of $|a|$:** Larger means _____, smaller means _____

Part C: Vertical Shifts (8 problems)

** Explore:** Graph these in GeoGebra: - $y = x^2$ (red) - $y = x^2 + 3$ (blue) - $y = x^2 - 2$ (green)

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17. Complete the table:

x	-2	-1	0	1	2
$f(x) = x^2$					
$g(x) = x^2 + 3$					
$h(x) = x^2 - 2$					

18. How does $y = x^2 + 3$ compare to $y = x^2$?

19. How does $y = x^2 - 2$ compare to $y = x^2$?

20. For each function, state the vertex: - (a) $y = x^2 \rightarrow$ Vertex: (___, ___) - (b) $y = x^2 + 5 \rightarrow$ Vertex: (___, ___) - (c) $y = x^2 - 4 \rightarrow$ Vertex: (___, ___) - (d) $y = x^2 + 100 \rightarrow$ Vertex: (___, ___)

21. The equation $y = x^2 + k$ shifts the parabola: - Up by k units if k is _____ - Down by $|k|$ units if k is _____

22. Write the equation of a parabola that: - (a) Has the same shape as $y = x^2$ but vertex at $(0, 7)$ - (b) Has the same shape as $y = x^2$ but vertex at $(0, -3)$

23. How many x-intercepts (roots) does each have? - (a) $y = x^2 + 1$ - (b) $y = x^2$ - (c) $y = x^2 - 4$

24. Connect to algebra: Solve $x^2 - 4 = 0$. How do the solutions relate to the graph of $y = x^2 - 4$?

Part D: Horizontal Shifts (8 problems)

** Explore:** Graph these in GeoGebra: - $y = x^2$ (red) - $y = (x - 2)^2$ (blue) - $y = (x + 3)^2$ (green)

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25. Complete the table for $f(x) = (x - 2)^2$:

x	0	1	2	3	4
$(x - 2)$					
$f(x) = (x - 2)^2$					

26. What is the vertex of $y = (x - 2)^2$?

27. Complete the table for $g(x) = (x + 3)^2$:

x	-5	-4	-3	-2	-1
$(x + 3)$					
$g(x) = (x + 3)^2$					

28. What is the vertex of $y = (x + 3)^2$?

29. The Tricky Part: - $y = (x-2)^2$ shifts the parabola _____ (left/right) by 2 units - $y = (x+3)^2$ shifts the parabola _____ (left/right) by 3 units

The shift is OPPOSITE to the sign inside the parentheses!

30. State the vertex of each: - (a) $y = (x - 5)^2 \rightarrow$ Vertex: (___, ___) - (b) $y = (x + 1)^2 \rightarrow$ Vertex: (___, ___) - (c) $y = (x - 10)^2 \rightarrow$ Vertex: (___, ___) - (d) $y = (x + 7)^2 \rightarrow$ Vertex: (___, ___)

31. Write the equation of a parabola with: - (a) Vertex at $(4, 0)$, same shape as $y = x^2$ - (b) Vertex at $(-2, 0)$, same shape as $y = x^2$

32. The equation $y = (x - h)^2$ has vertex at $(h, 0)$.

If $h = 3$, the vertex is at _____. If $h = -4$, the vertex is at _____.

Part E: Vertex Form — Putting It All Together (10 problems)

The **vertex form** of a quadratic is:

$$y = a(x - h)^2 + k$$

Where: - (h, k) is the **vertex** - a determines **direction** (up/down) and **width**

** Explore:** Vertex Form Explorer *Use sliders to change a, h, and k and watch the parabola move!*

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33. For $y = 2(x - 1)^2 + 3$: - (a) What is a? _____ - (b) What is h? _____ - (c) What is k? _____ - (d) What is the vertex? (___, ___) - (e) Does it open up or down? _____ - (f) Is it narrower or wider than $y = x^2$? _____

34. For $y = -0.5(x + 2)^2 - 1$: - (a) What is a? _____ - (b) What is h? _____ *(Careful with the sign!)* - (c) What is k? _____ - (d) What is the vertex? (___, ___) - (e) Does it open up or down? _____ - (f) Is it narrower or wider than $y = x^2$? _____

35. Write the equation in vertex form for a parabola with: - (a) Vertex $(3, 2)$, opens up, same width as $y = x^2$ - (b) Vertex $(-1, 4)$, opens down, same width as $y = x^2$ - (c) Vertex $(0, -5)$, opens up, twice as narrow as $y = x^2$ - (d) Vertex $(2, -3)$, opens down, half as steep as $y = x^2$

36. Match each equation to its vertex:

Equation	Vertex
$y = (x - 4)^2 + 1$	A. $(4, 1)$
$y = (x + 4)^2 - 1$	B. $(-4, -1)$
$y = (x - 4)^2 - 1$	C. $(4, -1)$
$y = (x + 4)^2 + 1$	D. $(-4, 1)$

37. Sketch (without GeoGebra) the graph of $y = (x-2)^2 - 4$: - (a) Vertex: _____ - (b) Opens: _____ - (c) y-intercept (plug in $x = 0$): _____ - (d) x-intercepts (set $y = 0$ and solve): _____

38. Sketch $y = -(x+1)^2 + 9$: - (a) Vertex: _____ - (b) Opens: _____ - (c) y-intercept: _____ - (d) x-intercepts: _____

39. A parabola has vertex $(2, 5)$ and passes through $(0, 1)$. Find its equation.

Hint: Start with $y = a(x-2)^2 + 5$ and use the point to find a.

40. A parabola has vertex $(-1, -4)$ and passes through $(1, 0)$. Find its equation.

41. Maximum/Minimum Values:

For $y = a(x-h)^2 + k$: - If $a > 0$, the parabola has a _____ value of _____ at $x =$ _____ - If $a < 0$, the parabola has a _____ value of _____ at $x =$ _____

42. A ball's height is $h(t) = -5(t-3)^2 + 50$. - (a) What is the maximum height? - (b) When is the maximum height reached? - (c) What was the initial height ($t = 0$)?

Part F: Standard Form and Converting (10 problems)

The **standard form** is:

$$y = ax^2 + bx + c$$

To find the vertex from standard form:

$$x_{\text{vertex}} = \frac{-b}{2a}$$

Then plug this x back in to find y_{vertex}.

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43. For $y = x^2 - 6x + 5$: - (a) Identify a, b, c: _____, _____, _____ - (b) Find $x_{\text{vertex}} = \frac{-b}{2a}$: _____ - (c) Find y_{vertex} by plugging in: _____ - (d) Vertex: (___, ___)

44. For $y = x^2 + 4x + 3$: - (a) Find the vertex. - (b) Find the y-intercept. - (c) Find the x-intercepts by factoring.

45. For $y = -x^2 + 2x + 8$: - (a) Find the vertex. - (b) Does this parabola have a maximum or minimum? - (c) Find the x-intercepts.

46. For $y = 2x^2 + 8x + 6$: - (a) Find the vertex. - (b) Factor to find x-intercepts.

47. Converting Standard to Vertex Form

To convert $y = x^2 - 4x + 7$ to vertex form, complete the square:

$y = x^2 - 4x + 7$ $y = (x^2 - 4x + 4) + 7 - 4 \leftarrow$ Add and subtract $(4/2)^2 = 4$ $y = (x-2)^2 + 3$

Convert to vertex form: - (a) $y = x^2 + 6x + 10$ - (b) $y = x^2 - 2x - 3$

48. Convert to vertex form: - (a) $y = x^2 + 8x + 20$ - (b) $y = x^2 - 10x + 21$

49. Converting Vertex to Standard Form

Expand $y = 2(x-3)^2 + 1$:

$y = 2(x^2 - 6x + 9) + 1 = 2x^2 - 12x + 18 + 1 = 2x^2 - 12x + 19$

Convert to standard form: - (a) $y = (x + 2)^2 - 5$ - (b) $y = 3(x - 1)^2 + 4$

50. Convert to standard form: - (a) $y = -(x - 4)^2 + 10$ - (b) $y = 2(x + 3)^2 - 7$

Part G: Roots and the Discriminant — Graphical View (10 problems)

The **x-intercepts** (roots) of a parabola are where $y = 0$.

The **discriminant** $b^2 - 4ac$ tells us how many roots to expect:

Discriminant	Roots	Graph
$b^2 - 4ac > 0$	Two distinct real roots	Crosses x-axis twice
$b^2 - 4ac = 0$	One repeated root	Touches x-axis once (vertex on x-axis)
$b^2 - 4ac < 0$	No real roots	Never crosses x-axis

** Explore:** Discriminant Visualizer

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51. For each equation, calculate the discriminant and predict the number of x-intercepts:

- (a) $y = x^2 - 4x + 3 \rightarrow$ Discriminant: _____, x-intercepts: _____
- (b) $y = x^2 - 4x + 4 \rightarrow$ Discriminant: _____, x-intercepts: _____
- (c) $y = x^2 - 4x + 5 \rightarrow$ Discriminant: _____, x-intercepts: _____

52. Verify your predictions from problem 51 by: - (a) Factoring or using quadratic formula for each - (b) Sketching each parabola (they all have the same vertex x-coordinate!)

53. A parabola has equation $y = x^2 + bx + 9$. For what values of b does it: - (a) Cross the x-axis twice? - (b) Touch the x-axis exactly once? - (c) Never touch the x-axis?

54. The graph of $y = x^2 - 6x + k$ will touch the x-axis exactly once when $k =$ _____.

55. For $y = ax^2 + 4x + 1$: - (a) If $a = 1$, how many x-intercepts? - (b) If $a = 4$, how many x-intercepts? - (c) If $a = 5$, how many x-intercepts?

56. Connecting Algebra and Geometry:

The equation $x^2 + 2x + 5 = 0$ has no real solutions.

Graphically, this means the parabola $y = x^2 + 2x + 5$ _____.

Find the vertex to confirm it's entirely above the x-axis.

57. Sketch a parabola that: - (a) Opens upward and has two x-intercepts - (b) Opens downward and has two x-intercepts - (c) Opens upward and has no x-intercepts - (d) Opens downward and has no x-intercepts

58. A parabola has vertex $(3, -4)$ and $a = 1$. - (a) Write its equation. - (b) Will it cross the x-axis? Why? - (c) Find the x-intercepts.

59. A parabola has vertex $(2, 5)$ and $a = 1$. - (a) Write its equation. - (b) Will it cross the x-axis? Why?

60. A parabola has vertex $(-1, 3)$ and $a = -2$. - (a) Write its equation. - (b) Will it cross the x-axis? Why? - (c) Find the x-intercepts.

Part H: Real-World Graphing Applications (6 problems)

61. A ball is thrown with height $h(t) = -5t^2 + 20t + 1$.

- (a) Write in vertex form.
- (b) What is the maximum height?
- (c) When is maximum height reached?
- (d) Sketch the graph for $t \geq 0$.
- (e) When does the ball hit the ground?

62. A company's profit (in thousands) is modeled by $P(x) = -2x^2 + 24x - 54$ where x is units sold (in hundreds).

- (a) Find the vertex. What does it represent?
- (b) What sales level maximizes profit?
- (c) What is the maximum profit?
- (d) At what sales levels does the company break even ($P = 0$)?

63. A farmer has 200m of fencing for a rectangular enclosure against a barn (one side needs no fence). The area is:

$$A(w) = w(200 - 2w) = -2w^2 + 200w$$

- (a) What width maximizes the area?
- (b) What is the maximum area?
- (c) Sketch $A(w)$ for sensible values of w.

64. The demand for a product is $D(p) = 1000 - 25p$ where p is price in euros.

Revenue is price $\times$ quantity: $R(p) = p \cdot D(p) = p(1000 - 25p) = -25p^2 + 1000p$

- (a) What price maximizes revenue?
- (b) What is the maximum revenue?
- (c) At what prices is revenue zero? Interpret these.

65. A suspension bridge cable forms a parabola. With the origin at the lowest point, the cable follows $y = 0.01x^2$ where x and y are in meters.

- (a) How high is the cable at $x = 50$m from center?
- (b) If the towers are 100m apart, how high are they above the lowest point?

- (c) Sketch the cable.

66. Water from a fountain follows $y = -0.5x^2 + 2x$ where x is horizontal distance and y is height (meters).

- (a) What is the maximum height of the water?
- (b) How far horizontally does the water travel?
- (c) Sketch the water's path.

GeoGebra Quick Reference

To graph in GeoGebra (geogebra.org/calculator):

1. Type equation directly: `y = x^2` or `y = 2(x-3)^2 + 1`
2. Use `^` for exponents: `x^2` means x^2
3. Use `*` for multiplication: `2*x^2` or just `2x^2`
4. Add sliders: Type `a = 1` then use a in equations

Useful explorations: - Type `y = a*x^2` and create slider for a - Type `y = (x - h)^2 + k` and create sliders for h and k - Type `y = a*(x - h)^2 + k` for full control